

Sample Pre-Lab Quiz Questions for Expt 4

Duration: 10 min; Max marks: 5 marks

Q1. With regard to the LED Blink program available in Arduino IDE for testing the ESP32-WROOM board, **mark all statements that are correct:**

- A) For the **Blink** program to work, an external LED should be connected to the ESP32-WROOM board.
- B) The **Blink** program does not require an external LED to be connected to the ESP32-WROOM board.
- C) The **Blink** program changes slowly and repetitively the brightness of the LED from the smallest value to the largest value and then back to the smallest value.
- D) The **Blink** program switches the LED ON for 500 ms and then OFF for 500 ms repetitively.
- E) The **Blink** program switches the LED ON for 1000 ms and then OFF for 1000 ms repetitively.

Q2 Mark all statements which are correct.

- A) ESP32-WROOM board has in-built 8-bit ADC for reading analog voltages.
- B) ESP32-WROOM board has in-built 10-bit ADC for reading analog voltages.
- C) ESP32-WROOM board has in-built 12-bit ADC for reading analog voltages.
- D) Output of the ADC can be used directly to control the PWM duty cycle.
- E) Output of the ADC should be mapped onto 8-bit range before using it as the PWM value.
- F) Output of the ADC should be mapped onto 10-bit range before using it as the PWM value.

Q3 Mark all statements which are correct.

- A) For generating analog voltages both Arduino Nano and ESP32-WROOM boards have in-built 8-bit DACs.
- B) For generating analog voltages both Arduino Nano and ESP32-WROOM boards have in-built 10-bit DACs.
- C) For generating analog voltages, the Arduino Nano board has an in-built 8-bit DAC whereas the ESP32-WROOM board has an in-built 10-bit DAC.
- D) For generating analog voltages, the Arduino Nano board has an in-built 10-bit DAC whereas the ESP32-WROOM board has an in-built 8-bit DAC.
- E) Arduino Nano board has no in-built DAC whereas the ESP32-WROOM board has an in-built DAC.
- F) Both Arduino Nano and ESP32-WROOM boards have no in-built DAC.

Q4. The following table gives a summary of the connections between the L298 motor driver card and the ESP32-WROOM board. With regard to these connections, **mark all statements that are correct.**

Links: L3 in; L1 removed.

Power:

Driver card	Power supply
V_S (10-12 V)	Supply+
Ground	Supply–
5 V Terminal	Open

Motor to Driver card

OUT1, OUT2 - Motor wires

Motor card to ESP32

Driver Card	ESP32
Ground	Gnd
Enable A	D18
IN1	D19
IN2	D23

- A) Assume that the connections mentioned above have been made correctly and the logical levels on IN1 and IN2 terminals were chosen to run the BO motor. If the Enable A terminal on the motor drive driver card is now made HIGH, the BO motor would start rotating.
- B) Assume that the connections mentioned above have been made correctly and the logical levels on IN1 and IN2 terminals were chosen to run the BO motor. If the Enable A terminal on the motor drive driver card is now made HIGH, the BO motor will not rotate.

- C) The IN1 and IN2 terminals of the motor driver are used by Arduino to control the direction of rotation of the BO motor.
- D) The IN1 and IN2 terminals of the motor driver card are used by Arduino to generate PWM waveforms to control the speed of the BO motor.
- E) The IN1 and IN2 terminals of the motor driver card are used by Arduino to run the BO motor by keeping IN1 = HIGH and by generating a PWM waveform on IN2.
- F) The IN1 and IN2 terminals of the motor driver card are used by Arduino to run the BO motor by keeping IN2 = HIGH and by generating a PWM waveform on IN1.

Q5. With regard to the BO motor and the L298 driver, **mark all statements that are correct.**

- A) The digital outputs of ESP32-WROOM board can be used directly to drive a BO motor.
- B) The digital outputs of ESP32-WROOM board cannot be used directly to drive a BO motor.
- C) Assume that proper connections between ESP32-WROOM and the motor driver card are made and the BO motor is presently running with a PWM value of 200. If the PWM value is changed from 200 to 150, the BO motor speed will increase.
- D) Assume that proper connections between ESP32-WROOM and the motor driver card are made and the BO motor is presently running with a PWM value of 200. If the PWM value is changed from 200 to 150, the BO motor speed will decrease.
- E) We put a 0.1 μ F capacitor across BO motor terminals to cancel the inductance of motor windings.
- F) We put a 0.1 μ F capacitor across BO motor terminals to increase its speed.
- G) We put a 0.1 μ F capacitor across BO motor terminals to suppress noise.
